Integer Programming Problems

Branch and Bound Method

Problems –

1] A furniture company manufactures dining tables and desks. Production of tables needs 2 units of labour hours and that of desk needs 5 units of labour hours. 25 hours of labour are available per week. No more than 8 dining tables and 4 desks can be sold in one week. Determine number of dining tables and desks to be manufactured, so that profit is maximized. Profit are obtained by selling 1 dining table is 3 units and 5 units of profit is obtained by selling one desk.
Answer: $Z_{max} = 31$ and tables = 7 and desks = 2.

2] A manufacturer of machine components makes two types of components - X and Y. The two components are processed on machines A and B. One unit of component X requires 2 hours on machine A and 6 hours on machine B. One unit of component Y requires 5 hours on machine A and 5 hours on machine B. There are 16 hours of time available on machine A and 30 hours on machine B. Profit gained by selling one unit of component X is Rs.10 and that of component Y is Rs.12. What should be the production of each of the two components so as to maximize the profit?
Answer: $x=3$, $y=2$, $Z_{max}=54$

3] The owner of a readymade garment store makes two types of shirts known as Zee shirts and Alpha shirts. He makes a profit of Rs.10 and Rs.40 per shirt on these two respectively. He has 2 tailors: A and B at his disposal to stitch these shirts. Tailor A and tailor B can devote utmost 7 hours and 15 hours per day respectively. Both these shirts are to be stitched by both the tailors. Tailor A and B spend 2 hours and 5 hours respectively in stitching Zee shirts and 4 hours and 3 hours respectively in stitching Alpha shirts. How many shirts of both the type should be stitched in order to maximize daily profits?
Answer: $x = 1$, $y = 1$, $Z_{max} = 50$.

4] Solve the following IPP using branch and bound method

Maximize $Z = 5x + 4y$

Subject to $x + y \le 5$

$10x + 6y \le 45$

$x, y \ge 0$ and integers.

$x = 3$, $y = 2$.

$Z_{max} = 23$

5] Solve the following IPP using branch and bound method

Maximize $Z = 15x_1 + 18x_2$

subject to $2x_1 + x_2 \le 20$

$x_1 + 2x_2 \le 24$

and $x_1, x_2 \ge 0$ and integers

Answer: $x = 4$, $y = 10$, $Z_{max} = 240$

Gomory Cut Method

1] Arts student paints wall portraits of two different sizes. The profit earned from the sale of larger size portrait and that from the smaller size portrait are Rs.20 and Rs.30. She can sell at the most 3 larger portraits and 5 smaller portraits. Because examinations are approaching fast she can devote at the most 10 hours during the week with each large portrait requiring 2 hours of work and small one taking 1 hour. If she does not wish to keep any incomplete portrait at the end of the week, determine how many portraits of each size, should she paint to maximize the profit.

Maximize $Z = 20x + 30y$

Subject to $2x + y \le 10$

$x \le 3$

$y \le 5$

$x, y \ge 0$ and integers.

Answer: $Z_{max} = 190$ and large paintings = 2 and small paintings = 5.

2] Maximize $Z = 20x + 30y$

Subject to $2x + 4y \le 17$

$3x + 3y \le 15$

$x, y \ge 0$ and integers.

Answer: $x = 2, y = 3, Z_{max} = 130$

3] Maximize $Z = 15x + 18y$

subject to $2x + y \le 20$

$x + 2y \le 24$

and $x, y \ge 0$ and integers

Answer: $x = 4, y = 10, Z_{max} = 240$

4] Maximize $Z = 5x + 6y$

Subject to $2x + 5y \le 16$

$6x + 5y \le 30$

$x, y \ge 0$ and integers.

Answer: $x = 3, y = 2, Z_{max} = 27$

5] Maximize $Z = 10x + 12y$

Subject to $5x + 3y \le 40$

$4x + 5y \le 30$

$x, y \ge 0$ and integers.

Answer: $x = 5, y = 2, Z_{max} = 74$

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